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(54) **FUNGICIDAL COMBINATIONS**

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A61F 2/92 (2013.01)
A61F 2/95 (2013.01)

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See application file for complete search history.

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ABSTRACT

A combination including a dithiocarbamate fungicide, a succinate dehydrogenase fungicide and at least one of ergosterol biosynthesis inhibitor fungicide or a quinone outside inhibitor fungicide.

4 Claims, No Drawings

(56)

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FUNGICIDAL COMBINATIONS**CROSS REFERENCE TO RELATED APPLICATIONS**

This application is a Continuation of U.S. application Ser. No. 16/347,283, filed on May 3, 2019, which is a National Stage application of PCT/IB2017/056710, filed on Oct. 30, 2017, which claims the benefit of Indian Application No. 201631037704, filed on Nov. 4, 2016, both of which are incorporated by reference in their entirety herein.

TECHNICAL FIELD

The present invention relates to a combination of fungicides. More specifically, the present invention relates to fungicidal combinations comprising succinate dehydrogenase inhibitor fungicides for controlling a broad spectrum of fungal diseases.

BACKGROUND OF THE INVENTION

Fungicides are an integral and important tool yielded by farmers to control diseases, as well as to improve yields and quality of the crops. There are various fungicides that have been developed over the years with many desirable attributes such as specificity, systemicity, curative and eradicant action and high activity at low use rates.

Succinate dehydrogenase inhibitor (SDHI) fungicides are known in the art to be broad spectrum and have a high potency. Pyrazolecarboxamides are a group of active compounds within the SDHI family of fungicides that are known to be more potent than most other SDHI fungicides. These molecules specifically bind to the ubiquinone-binding site (Q-site) of the mitochondrial complex II, thereby inhibiting fungal respiration. These fungicides are known to control a broad spectrum of fungal diseases.

Various other classes of fungicides are also known in the art, such as Quinone outside inhibitors (Qols), ergosterol-biosynthesis inhibitors, fungicides that act on multiple sites, fungicides that affect mitosis etc. These fungicides have been mixed with SDHI fungicides to achieve a broad spectrum of disease control.

WO2006037632 teaches combinations of SDHI fungicides with a second active compound. WO2013127818 teaches combinations of SDHI fungicides with various herbicides. WO2006037634 teaches methods of controlling fungi using a combination of SDHI fungicide with various fungicides. However, the prior art does not teach the use of ternary or higher combinations of SDHI fungicides.

Dithiocarbamate are known in the art as multi-site fungicides. These fungicides are used for broad-spectrum disease control in more than 70 crops. Mancozeb is especially important for controlling devastating and fast spreading diseases such as *Phytophthora infestans*, *Venturia inaequalis* etc. Dithiocarbamate fungicides, especially mancozeb, are particularly useful for disease control because of their broad spectrum of activity, high tolerance by crop plants, and general usefulness for controlling fungal plant diseases not controlled by active compounds that act on only a single target site in the fungus.

Mancozeb has been combined with various SDHI fungicides for disease control. There is a need in the art to improve on the disease spectrum provided by these combinations.

There is therefore a need in the art for combinations of SDHIs with a specific fungicide that helps improve spec-

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trum. With crop tolerances decreasing, lower use rates being imposed and resistance being increasingly observed, there is a need for a combination of actives that allows for broader disease control spectrum that combines curative and preventive actives and has a lower dosage.

Therefore, embodiments of the present invention may ameliorate one or more of the above mentioned problems.

Therefore, embodiments of the present invention may provide combinations of fungicides that possess an enhanced efficacy over the individual fungicides used in isolation.

Another object of the present invention is to provide a fungicidal combination that causes an enhanced greening of the crops to which it is administered.

Another object of the present invention is to provide a fungicidal combination that causes late senescence to the crop to which it is applied thereby resulting into an increasing yield of the crop.

Yet another object of the present invention is to provide a fungicidal combination that results into reduced fungal disease incidence in the crops to which it is applied.

Another object of the present invention is to provide a fungicidal combination that achieves increased yield in the crops to which it is applied.

Some or all these and other objects of the invention are can be achieved by way of the invention described herein-after.

SUMMARY OF THE INVENTION

Thus, an aspect of the present invention can provide a fungicidal combination comprising at least one succinate dehydrogenase inhibitor fungicide, at least one dithiocarbamate fungicide, and at least another fungicide.

Another aspect of the present invention can provide a fungicidal combination comprising at least one succinate dehydrogenase inhibitor fungicide, at least one dithiocarbamate fungicide and at least two other fungicides.

Another aspect of the present invention can provide synergistic compositions comprising at least one succinate dehydrogenase inhibitor fungicide, at least one dithiocarbamate fungicide, and at least one other fungicide.

Another aspect of the present invention can provide synergistic compositions comprising at least one succinate dehydrogenase inhibitor fungicide, at least one dithiocarbamate fungicide and at least two other fungicides.

DETAILED DESCRIPTION

The term 'disease control' as used herein denotes control and prevention of a disease. Controlling effects include all deviation from natural development, for example: killing, retardation, decrease of the fungal disease. The term 'plants' refers to all physical parts of a plant, including seeds, seedlings, saplings, roots, tubers, stems, stalks, foliage and fruits. The term "locus" of a plant as used herein is intended to embrace the place on which the plants are growing, where the plant propagation materials of the plants are sown or where the plant propagation materials of the plants will be placed into the soil. The term "plant propagation material" is understood to denote generative parts of a plant, such as seeds, vegetative material such as cuttings or tubers, roots, fruits, tubers, bulbs, rhizomes and parts of plants, germinated plants and young plants which are to be transplanted after germination or after emergence from the soil. These young plants may be protected before transplantation by a total or partial treatment by immersion. The term "agricul-

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turally acceptable amount of active" refers to an amount of an active that kills or inhibits the plant disease for which control is desired, in an amount not significantly toxic to the plant being treated.

Succinate dehydrogenase inhibitor (SDHI) fungicides play an important role in plant protection against many phytopathogenic fungi. These molecules specifically bind to the ubiquinone-binding site (Q-site) of the mitochondrial complex II, thereby inhibiting fungal respiration. Dithiocarbamate are multi-site contact fungicides. These molecules

attack multiple sites within the fungal cells. It has surprisingly been found that the addition of a dithiocarbamate fungicide to the combinations of succinate dehydrogenase inhibitors with at least another fungicide selected from an ergosterol biosynthesis inhibitors or Quinone outside inhibitors or a combination thereof, resulted in surprising and unexpected advantages. It was surprising that the addition of a dithiocarbamate fungicide to the combination of a succinate dehydrogenase inhibitor with at least another fungicide selected from an ergosterol biosynthesis inhibitors and/or Quinone outside inhibitors or a combination thereof resulted in an enhancement of the efficacy, and a surprising reduction in fungal disease incidence, seen only with the combination of succinate dehydrogenase inhibitors with at least another fungicide selected from an ergosterol biosynthesis inhibitors or Quinone outside inhibitors or a combination thereof. It has further been found that the addition of a dithiocarbamate fungicide to these combinations and application of these combinations during the flowering stage of the crop delayed the senescence in the crop to which they were applied, which led to better greening in the crop thereby increasing the level of photosynthesis occurring within the plant, thereby leading to a greater yield from the crop to which they were applied.

These surprising advantages of the combinations of the invention were not observed when the dithiocarbamate fungicide was not present in the combination. Therefore, these unexpected advantages of the combination of the present invention could be attributed to the inclusion of a dithiocarbamate fungicide to the combination of a succinate dehydrogenase inhibitor with at least another fungicide selected from an ergosterol biosynthesis inhibitors and/or Quinone outside inhibitors or a combination thereof.

Thus, in an aspect, the present invention provides a fungicidal combination comprising:

- (a) at least one dithiocarbamate fungicide;
- (b) at least one succinate dehydrogenase inhibitor fungicide; and
- (c) at least another fungicide selected from an ergosterol biosynthesis inhibitors and/or Quinone outside inhibitors.

In an embodiment, the dithiocarbamate fungicide is selected from the group consisting of amobam, asomate, azithiram, carbamorph, cufraneb, cuprobam, disulfiram, ferbam, metam, nabam, tecoram, thiram, urbacide, ziram, dazomet, etem, milneb, mancopper, mancozeb, maneb, metiram, polycarbamate, propineb and zineb.

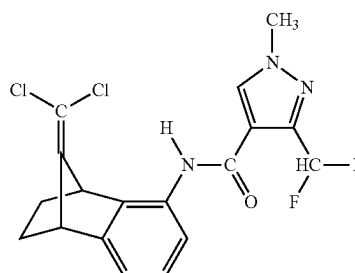
In an embodiment, the preferred dithiocarbamate fungicide is mancozeb.

In an embodiment, the succinate dehydrogenase inhibitor is selected from pyrazole carboxamide class of succinate dehydrogenase inhibitor fungicides. However, it should be understood that the choice of succinate dehydrogenase inhibitors is not understood to be limited to these pyrazole carboxamide fungicides alone.

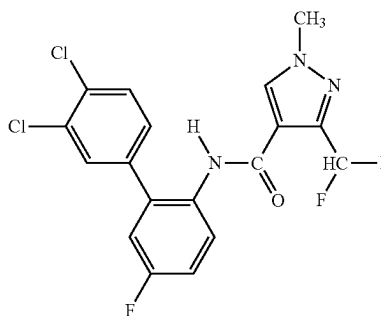
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In an embodiment, the pyrazole carboxamide class of succinate dehydrogenase inhibitor fungicide may be selected from benzovindiflupyr, bixafen, fluxapyroxad, furametpyr, isopyrazam, penflufen, penthiopyrad, and sedaxane.

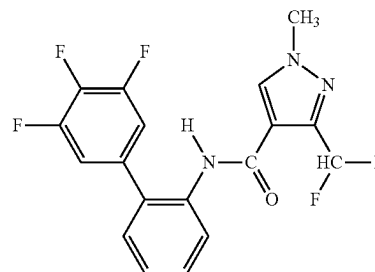
Benzovindiflupyr has the chemical name N-[(1RS,4SR)-9-(dichloromethylene)-1,2,3,4-tetrahydro-1,4-methanonaphthalen-5-yl]-3-(difluoromethyl)-1-methylpyrazole-4-carboxamide and has the structure:



Bixafen has the chemical name N-(3',4'-dichloro-5-fluorobiphenyl-2-yl)-3-(difluoromethyl)-1-methylpyrazole-4-carboxamide and the structure:

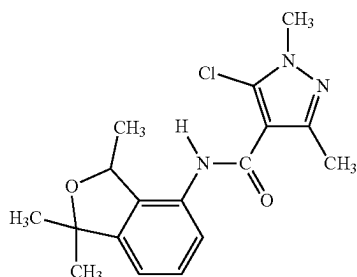


Fluxapyroxad has the chemical name 3-(difluoromethyl)-1-methyl-N-(3',4',5'-trifluorobiphenyl-2-yl)pyrazole-4-carboxamide and has the structure:

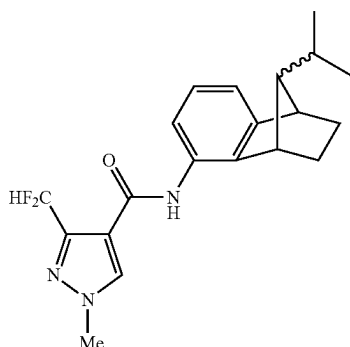


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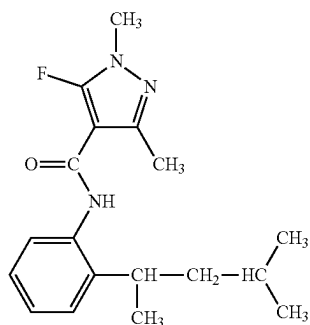
Furametypr has the chemical name (RS)-5-chloro-N-(1,3-dihydro-1,1,3-trimethylisobenzofuran-4-yl)-1,3-dimethylpyrazole-4-carboxamide and has the structure:



Isopyrazam is a mixture of 2 isomers 3-(difluoromethyl)-1-methyl-N-[(1RS,4SR,9RS)-1,2,3,4-tetrahydro-9-isopropyl-1,4-methanonaphthalen-5-yl]pyrazole-4-carboxamide and 2 isomers 3-(difluoromethyl)-1-methyl-N-[(1RS,4SR,9SR)-1,2,3,4-tetrahydro-9-isopropyl-1,4-methanonaphthalen-5-yl]pyrazole-4-carboxamide and its tautomer's have the structure:

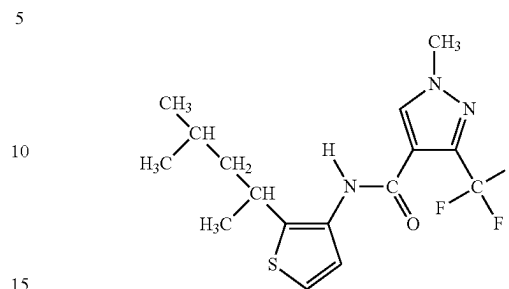


Penflufen has the chemical name N-[2-(1,3-dimethylbutyl)phenyl]-5-fluoro-1,3-dimethyl-1H-pyrazole-4-carboxamide, and has the following structure:

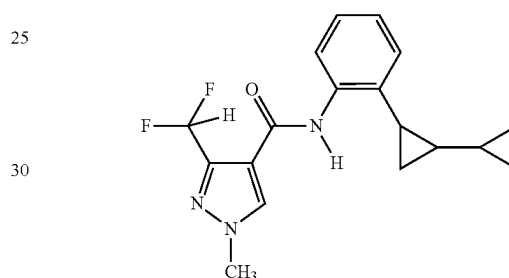


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Penthiopyrad has the chemical name (RS)—N-[2-(1,3-dimethylbutyl)-3-thienyl]-1-methyl-3-(trifluoromethyl)pyrazole-4-carboxamide, and has the following structure:



Sedaxane is a mixture of 2 cis-isomers 2'-[(1RS,2RS)-1,1'-bicycloprop-2-yl]-3-(difluoromethyl)-1-methylpyrazole-4-carboxanilide and 2 trans-isomers 2'-[(1RS,2SR)-1,1'-bicycloprop-2-yl]-3-(difluoromethyl)-1-methylpyrazole-4-carboxanilide, and its tautomer's have the structure:



In an embodiment, the succinate dehydrogenase inhibitor fungicide may be selected from the group consisting of benodanil, flutolanil, mepronil, isofetamid, fluopyram, fenfuram, carboxin, oxycarboxin, thifluzamide, boscalid and IR9792.

In an embodiment, the third and/or fourth fungicides in the combinations of the present invention may be selected from ergosterol biosynthesis inhibitors, and/or Quinone outside (Qo) inhibitors or mixtures thereof.

Thus in an embodiment, the ergosterol biosynthesis inhibitors may be selected from the group consisting of azaconazole, bitertanol, bromuconazole, cyproconazole, difenoconazole, diniconazole, epoxiconazole, etaconazole, fenbuconazole, fluquinconazole, flusilazole, flutriafol, hexaconazole, imibenconazole, Ipconazole, metconazole, myclobutanil, penconazole, Propiconazole, simeconazole, tebuconazole, tetraconazole, triadimefon, triadimenol, triticonazole, prothioconazole, imazalil, oxpoconazole, pefurazoate, prochloraz, triflumizole, fenarimol, nuarimol, pyrifenoxy, pyrisoxazole, triforine and mixtures thereof.

In another embodiment, the ergosterol biosynthesis inhibitors may be selected from prothioconazole, tebuconazole, hexaconazole, cyroconazole or epoxiconazole.

In an embodiment, the third fungicide may be a Quinone outside (Qo) inhibitor fungicide selected from azoxystrobin, coumoxystrobin, enoxystrobin, flufenoxystrobin, picoxystrobin, pyraoxystrobin, mandestrobin, pyraclostrobin, pyrametostrobin, triclopyricarb, kresoxim-methyl, dimoxystrobin, fenaminostrobin, metominostrobin, trifloxystrobin, famoxadone, fluoxastrobin, fenamidone, pyribencarb and mixtures thereof.

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In an embodiment, the Quinone outside (Qo) inhibitor fungicide may be selected from azoxystrobin, picoxystrobin, kresoxim-methyl, pyraclostrobin and trifloxystrobin.

In an embodiment of the combinations of the present invention, the preferred succinate dehydrogenase inhibitor fungicide is isopyrazam.

In an embodiment, the combinations of the present invention include the following preferred combinations:

S No.	I	II	III	IV
1	Mancozeb	Isopyrazam	Cyproconazole	—
2	Mancozeb	Isopyrazam	Difenoconazole	—
3	Mancozeb	Isopyrazam	Epoxiconazole	—
4	Mancozeb	Isopyrazam	Hexaconazole	—
5	Mancozeb	Isopyrazam	Tebuconazole	—
6	Mancozeb	Isopyrazam	Tetraconazole	—
7	Mancozeb	Isopyrazam	Prothioconazole	—
8	Mancozeb	Isopyrazam	—	Azoxystrobin
9	Mancozeb	Isopyrazam	—	Picoxystrobin
10	Mancozeb	Isopyrazam	—	Pyraclostrobin
11	Mancozeb	Isopyrazam	—	Kresoxim-methyl
12	Mancozeb	Isopyrazam	—	Trifloxystrobin
13	Mancozeb	Isopyrazam	Cyproconazole	Azoxystrobin
14	Mancozeb	Isopyrazam	Cyproconazole	Picoxystrobin
15	Mancozeb	Isopyrazam	Cyproconazole	Pyraclostrobin
16	Mancozeb	Isopyrazam	Cyproconazole	Kresoxim-methyl
17	Mancozeb	Isopyrazam	Cyproconazole	Trifloxystrobin
18	Mancozeb	Isopyrazam	Difenoconazole	Azoxystrobin
19	Mancozeb	Isopyrazam	Difenoconazole	Picoxystrobin
20	Mancozeb	Isopyrazam	Difenoconazole	Pyraclostrobin
21	Mancozeb	Isopyrazam	Difenoconazole	Kresoxim-methyl
22	Mancozeb	Isopyrazam	Difenoconazole	Trifloxystrobin
23	Mancozeb	Isopyrazam	Epoxiconazole	Azoxystrobin
24	Mancozeb	Isopyrazam	Epoxiconazole	Picoxystrobin
25	Mancozeb	Isopyrazam	Epoxiconazole	Pyraclostrobin
26	Mancozeb	Isopyrazam	Epoxiconazole	Kresoxim-methyl
27	Mancozeb	Isopyrazam	Epoxiconazole	Trifloxystrobin
28	Mancozeb	Isopyrazam	Hexaconazole	Azoxystrobin
29	Mancozeb	Isopyrazam	Hexaconazole	Picoxystrobin
30	Mancozeb	Isopyrazam	Hexaconazole	Pyraclostrobin
31	Mancozeb	Isopyrazam	Hexaconazole	Kresoxim-methyl
32	Mancozeb	Isopyrazam	Hexaconazole	Trifloxystrobin
33	Mancozeb	Isopyrazam	Tebuconazole	Azoxystrobin
34	Mancozeb	Isopyrazam	Tebuconazole	Picoxystrobin
35	Mancozeb	Isopyrazam	Tebuconazole	Pyraclostrobin
36	Mancozeb	Isopyrazam	Tebuconazole	Kresoxim-methyl
37	Mancozeb	Isopyrazam	Tebuconazole	Trifloxystrobin
38	Mancozeb	Isopyrazam	Tetraconazole	Azoxystrobin
39	Mancozeb	Isopyrazam	Tetraconazole	Picoxystrobin
40	Mancozeb	Isopyrazam	Tetraconazole	Pyraclostrobin
41	Mancozeb	Isopyrazam	Tetraconazole	Kresoxim-methyl
42	Mancozeb	Isopyrazam	Tetraconazole	Trifloxystrobin
43	Mancozeb	Isopyrazam	Prothioconazole	Azoxystrobin
44	Mancozeb	Isopyrazam	Prothioconazole	Picoxystrobin
45	Mancozeb	Isopyrazam	Prothioconazole	Pyraclostrobin
46	Mancozeb	Isopyrazam	Prothioconazole	Kresoxim-methyl
47	Mancozeb	Isopyrazam	Prothioconazole	Trifloxystrobin

In an embodiment of the combinations of the present invention, the preferred succinate dehydrogenase inhibitor fungicide is Benzovindiflupyr.

In an embodiment, the combinations of the present invention include the following preferred combinations:

S No.	I	II	III	IV
48	Mancozeb	Benzovindiflupyr	Cyproconazole	—
49	Mancozeb	Benzovindiflupyr	Difenoconazole	—

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-continued

S No.	I	II	III	IV
50	Mancozeb	Benzovindiflupyr	Epoxiconazole	—
51	Mancozeb	Benzovindiflupyr	Hexaconazole	—
52	Mancozeb	Benzovindiflupyr	Tebuconazole	—
53	Mancozeb	Benzovindiflupyr	Tetraconazole	—
54	Mancozeb	Benzovindiflupyr	Prothioconazole	—
55	Mancozeb	Benzovindiflupyr	—	Azoxystrobin
56	Mancozeb	Benzovindiflupyr	—	Picoxystrobin
57	Mancozeb	Benzovindiflupyr	—	Pyraclostrobin
58	Mancozeb	Benzovindiflupyr	—	Kresoxim-methyl
59	Mancozeb	Benzovindiflupyr	—	Trifloxystrobin
60	Mancozeb	Benzovindiflupyr	Cyproconazole	Azoxystrobin
61	Mancozeb	Benzovindiflupyr	Cyproconazole	Picoxystrobin
62	Mancozeb	Benzovindiflupyr	Cyproconazole	Pyraclostrobin
63	Mancozeb	Benzovindiflupyr	Cyproconazole	Kresoxim-methyl
64	Mancozeb	Benzovindiflupyr	Cyproconazole	Trifloxystrobin
65	Mancozeb	Benzovindiflupyr	Difenoconazole	Azoxystrobin
66	Mancozeb	Benzovindiflupyr	Difenoconazole	Picoxystrobin
67	Mancozeb	Benzovindiflupyr	Difenoconazole	Pyraclostrobin
68	Mancozeb	Benzovindiflupyr	Difenoconazole	Kresoxim-methyl
69	Mancozeb	Benzovindiflupyr	Difenoconazole	Trifloxystrobin
70	Mancozeb	Benzovindiflupyr	Epoxiconazole	Azoxystrobin
71	Mancozeb	Benzovindiflupyr	Epoxiconazole	Picoxystrobin
72	Mancozeb	Benzovindiflupyr	Epoxiconazole	Pyraclostrobin
73	Mancozeb	Benzovindiflupyr	Epoxiconazole	Kresoxim-methyl
74	Mancozeb	Benzovindiflupyr	Epoxiconazole	Trifloxystrobin
75	Mancozeb	Benzovindiflupyr	Hexaconazole	Azoxystrobin
76	Mancozeb	Benzovindiflupyr	Hexaconazole	Picoxystrobin
77	Mancozeb	Benzovindiflupyr	Hexaconazole	Pyraclostrobin
78	Mancozeb	Benzovindiflupyr	Hexaconazole	Kresoxim-methyl
79	Mancozeb	Benzovindiflupyr	Hexaconazole	Trifloxystrobin
80	Mancozeb	Benzovindiflupyr	Tebuconazole	Azoxystrobin
81	Mancozeb	Benzovindiflupyr	Tebuconazole	Picoxystrobin
82	Mancozeb	Benzovindiflupyr	Tebuconazole	Pyraclostrobin
83	Mancozeb	Benzovindiflupyr	Tebuconazole	Kresoxim-methyl
84	Mancozeb	Benzovindiflupyr	Tebuconazole	Trifloxystrobin
85	Mancozeb	Benzovindiflupyr	Tetraconazole	Azoxystrobin
86	Mancozeb	Benzovindiflupyr	Tetraconazole	Picoxystrobin
87	Mancozeb	Benzovindiflupyr	Tetraconazole	Pyraclostrobin
88	Mancozeb	Benzovindiflupyr	Tetraconazole	Kresoxim-methyl
89	Mancozeb	Benzovindiflupyr	Tetraconazole	Trifloxystrobin
90	Mancozeb	Benzovindiflupyr	Prothioconazole	Azoxystrobin
91	Mancozeb	Benzovindiflupyr	Prothioconazole	Picoxystrobin
92	Mancozeb	Benzovindiflupyr	Prothioconazole	Pyraclostrobin
93	Mancozeb	Benzovindiflupyr	Prothioconazole	Kresoxim-methyl
94	Mancozeb	Benzovindiflupyr	Prothioconazole	Trifloxystrobin

In an embodiment of the combinations of the present invention, the preferred succinate dehydrogenase inhibitor fungicide is Penthiopyrad.

In an embodiment, the combinations of the present invention include the following preferred combinations:

S No.	I	II	III	IV
95	Mancozeb	Penthiopyrad	Cyproconazole	—
96	Mancozeb	Penthiopyrad	Difenoconazole	—
97	Mancozeb	Penthiopyrad	Epoxiconazole	—
98	Mancozeb	Penthiopyrad	Hexaconazole	—
99	Mancozeb	Penthiopyrad	Tebuconazole	—
100	Mancozeb	Penthiopyrad	Tetraconazole	—
101	Mancozeb	Penthiopyrad	Prothioconazole	—
102	Mancozeb	Penthiopyrad	—	Azoxystrobin
103	Mancozeb	Penthiopyrad	—	Picoxystrobin
104	Mancozeb	Penthiopyrad	—	Pyraclostrobin

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-continued

S No.	I	II	III	IV
105	Mancozeb	Penthiopyrad	—	Kresoxim-methyl
106	Mancozeb	Penthiopyrad	—	Trifloxystrobin
107	Mancozeb	Penthiopyrad	Cyproconazole	Azoxystrobin
108	Mancozeb	Penthiopyrad	Cyproconazole	Picoxystrobin
109	Mancozeb	Penthiopyrad	Cyproconazole	Pyraclostrobin
110	Mancozeb	Penthiopyrad	Cyproconazole	Kresoxim-methyl
111	Mancozeb	Penthiopyrad	Cyproconazole	Trifloxystrobin
112	Mancozeb	Penthiopyrad	Difenoconazole	Azoxystrobin
113	Mancozeb	Penthiopyrad	Difenoconazole	Picoxystrobin
114	Mancozeb	Penthiopyrad	Difenoconazole	Pyraclostrobin
115	Mancozeb	Penthiopyrad	Difenoconazole	Kresoxim-methyl
116	Mancozeb	Penthiopyrad	Difenoconazole	Trifloxystrobin
117	Mancozeb	Penthiopyrad	Epoxiconazole	Azoxystrobin
118	Mancozeb	Penthiopyrad	Epoxiconazole	Picoxystrobin
119	Mancozeb	Penthiopyrad	Epoxiconazole	Pyraclostrobin
120	Mancozeb	Penthiopyrad	Epoxiconazole	Kresoxim-methyl
121	Mancozeb	Penthiopyrad	Epoxiconazole	Trifloxystrobin
122	Mancozeb	Penthiopyrad	Hexaconazole	Azoxystrobin
123	Mancozeb	Penthiopyrad	Hexaconazole	Picoxystrobin
124	Mancozeb	Penthiopyrad	Hexaconazole	Pyraclostrobin
125	Mancozeb	Penthiopyrad	Hexaconazole	Kresoxim-methyl
126	Mancozeb	Penthiopyrad	Hexaconazole	Trifloxystrobin
127	Mancozeb	Penthiopyrad	Tebuconazole	Azoxystrobin
128	Mancozeb	Penthiopyrad	Tebuconazole	Picoxystrobin
129	Mancozeb	Penthiopyrad	Tebuconazole	Pyraclostrobin
130	Mancozeb	Penthiopyrad	Tebuconazole	Kresoxim-methyl
131	Mancozeb	Penthiopyrad	Tebuconazole	Trifloxystrobin
132	Mancozeb	Penthiopyrad	Tetraconazole	Azoxystrobin
133	Mancozeb	Penthiopyrad	Tetraconazole	Picoxystrobin
134	Mancozeb	Penthiopyrad	Tetraconazole	Pyraclostrobin
135	Mancozeb	Penthiopyrad	Tetraconazole	Kresoxim-methyl
136	Mancozeb	Penthiopyrad	Tetraconazole	Trifloxystrobin
137	Mancozeb	Penthiopyrad	Prothioconazole	Azoxystrobin
138	Mancozeb	Penthiopyrad	Prothioconazole	Picoxystrobin
139	Mancozeb	Penthiopyrad	Prothioconazole	Pyraclostrobin
140	Mancozeb	Penthiopyrad	Prothioconazole	Kresoxim-methyl
141	Mancozeb	Penthiopyrad	Prothioconazole	Trifloxystrobin

In an embodiment of the combinations of the present invention, the preferred succinate dehydrogenase inhibitor fungicide is boscalid.

In an embodiment, the combinations of the present invention include the following preferred combinations:

S No.	I	II	III	IV
142	Mancozeb	Boscalid	Cyproconazole	—
143	Mancozeb	Boscalid	Difenoconazole	—
144	Mancozeb	Boscalid	Epoxiconazole	—
145	Mancozeb	Boscalid	Hexaconazole	—
146	Mancozeb	Boscalid	Tebuconazole	—
147	Mancozeb	Boscalid	Tetraconazole	—
148	Mancozeb	Boscalid	Prothioconazole	—
149	Mancozeb	Boscalid	—	Azoxystrobin
150	Mancozeb	Boscalid	—	Picoxystrobin
151	Mancozeb	Boscalid	—	Pyraclostrobin
152	Mancozeb	Boscalid	—	Kresoxim-methyl
153	Mancozeb	Boscalid	—	Trifloxystrobin
154	Mancozeb	Boscalid	Cyproconazole	Azoxystrobin
155	Mancozeb	Boscalid	Cyproconazole	Picoxystrobin
156	Mancozeb	Boscalid	Cyproconazole	Pyraclostrobin
157	Mancozeb	Boscalid	Cyproconazole	Kresoxim-methyl
158	Mancozeb	Boscalid	Cyproconazole	Trifloxystrobin
159	Mancozeb	Boscalid	Difenoconazole	Azoxystrobin

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S No.	I	II	III	IV
160	Mancozeb	Boscalid	Difenoconazole	Picoxystrobin
161	Mancozeb	Boscalid	Difenoconazole	Pyraclostrobin
162	Mancozeb	Boscalid	Difenoconazole	Kresoxim-methyl
163	Mancozeb	Boscalid	Difenoconazole	Trifloxystrobin
164	Mancozeb	Boscalid	Epoxiconazole	Azoxystrobin
165	Mancozeb	Boscalid	Epoxiconazole	Picoxystrobin
166	Mancozeb	Boscalid	Epoxiconazole	Pyraclostrobin
167	Mancozeb	Boscalid	Epoxiconazole	Kresoxim-methyl
168	Mancozeb	Boscalid	Epoxiconazole	Trifloxystrobin
169	Mancozeb	Boscalid	Hexaconazole	Azoxystrobin
170	Mancozeb	Boscalid	Hexaconazole	Picoxystrobin
171	Mancozeb	Boscalid	Hexaconazole	Pyraclostrobin
172	Mancozeb	Boscalid	Hexaconazole	Kresoxim-methyl
173	Mancozeb	Boscalid	Hexaconazole	Trifloxystrobin
174	Mancozeb	Boscalid	Tebuconazole	Azoxystrobin
175	Mancozeb	Boscalid	Tebuconazole	Picoxystrobin
176	Mancozeb	Boscalid	Tebuconazole	Pyraclostrobin
177	Mancozeb	Boscalid	Tebuconazole	Kresoxim-methyl
178	Mancozeb	Boscalid	Tebuconazole	Trifloxystrobin
179	Mancozeb	Boscalid	Tetraconazole	Azoxystrobin
180	Mancozeb	Boscalid	Tetraconazole	Picoxystrobin
181	Mancozeb	Boscalid	Tetraconazole	Pyraclostrobin
182	Mancozeb	Boscalid	Tetraconazole	Kresoxim-methyl
183	Mancozeb	Boscalid	Tetraconazole	Trifloxystrobin
184	Mancozeb	Boscalid	Prothioconazole	Azoxystrobin
185	Mancozeb	Boscalid	Prothioconazole	Picoxystrobin
186	Mancozeb	Boscalid	Prothioconazole	Pyraclostrobin
187	Mancozeb	Boscalid	Prothioconazole	Kresoxim-methyl
188	Mancozeb	Boscalid	Prothioconazole	Trifloxystrobin

In an embodiment of the combinations of the present invention, the preferred succinate dehydrogenase inhibitor fungicide is IR9792. IR9792 is a SDHI fungicide, developed by Isagro, with the proposed common name fluindapyr.

In an embodiment, the combinations of the present invention include the following preferred combinations:

S No.	I	II	III	IV
189	Mancozeb	IR9792	Cyproconazole	—
190	Mancozeb	IR9792	Difenoconazole	—
191	Mancozeb	IR9792	Epoxiconazole	—
192	Mancozeb	IR9792	Hexaconazole	—
193	Mancozeb	IR9792	Tebuconazole	—
194	Mancozeb	IR9792	Tetraconazole	—
195	Mancozeb	IR9792	Prothioconazole	—
196	Mancozeb	IR9792	—	Azoxystrobin
197	Mancozeb	IR9792	—	Picoxystrobin
198	Mancozeb	IR9792	—	Pyraclostrobin
199	Mancozeb	IR9792	—	Kresoxim-methyl
200	Mancozeb	IR9792	—	Trifloxystrobin
201	Mancozeb	IR9792	Cyproconazole	Azoxystrobin
202	Mancozeb	IR9792	Cyproconazole	Picoxystrobin
203	Mancozeb	IR9792	Cyproconazole	Pyraclostrobin
204	Mancozeb	IR9792	Cyproconazole	Kresoxim-methyl
205	Mancozeb	IR9792	Cyproconazole	Trifloxystrobin
206	Mancozeb	IR9792	Difenoconazole	Azoxystrobin
207	Mancozeb	IR9792	Difenoconazole	Picoxystrobin
208	Mancozeb	IR9792	Difenoconazole	Pyraclostrobin
209	Mancozeb	IR9792	Difenoconazole	Kresoxim-methyl
210	Mancozeb	IR9792	Difenoconazole	Trifloxystrobin
211	Mancozeb	IR9792	Epoxiconazole	Azoxystrobin
212	Mancozeb	IR9792	Epoxiconazole	Picoxystrobin

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S No.	I	II	III	IV
213	Mancozeb	IR9792	Epoxiconazole	Pyraclostrobin
214	Mancozeb	IR9792	Epoxiconazole	Kresoxim-methyl
215	Mancozeb	IR9792	Epoxiconazole	Trifloxystrobin
216	Mancozeb	IR9792	Hexaconazole	Azoxystrobin
217	Mancozeb	IR9792	Hexaconazole	Picoxystrobin
218	Mancozeb	IR9792	Hexaconazole	Pyraclostrobin
219	Mancozeb	IR9792	Hexaconazole	Kresoxim-methyl
220	Mancozeb	IR9792	Hexaconazole	Trifloxystrobin
221	Mancozeb	IR9792	Tebuconazole	Azoxystrobin
222	Mancozeb	IR9792	Tebuconazole	Picoxystrobin
223	Mancozeb	IR9792	Tebuconazole	Pyraclostrobin
224	Mancozeb	IR9792	Tebuconazole	Kresoxim-methyl
225	Mancozeb	IR9792	Tebuconazole	Trifloxystrobin
226	Mancozeb	IR9792	Tetraconazole	Azoxystrobin
227	Mancozeb	IR9792	Tetraconazole	Picoxystrobin
228	Mancozeb	IR9792	Tetraconazole	Pyraclostrobin
229	Mancozeb	IR9792	Tetraconazole	Kresoxim-methyl
230	Mancozeb	IR9792	Tetraconazole	Trifloxystrobin
231	Mancozeb	IR9792	Prothioconazole	Azoxystrobin
232	Mancozeb	IR9792	Prothioconazole	Picoxystrobin
233	Mancozeb	IR9792	Prothioconazole	Pyraclostrobin
234	Mancozeb	IR9792	Prothioconazole	Kresoxim-methyl
235	Mancozeb	IR9792	Prothioconazole	Trifloxystrobin

In an embodiment of the combinations of the present invention, the preferred succinate dehydrogenase inhibitor fungicide is Bixafen.

In an embodiment, the combinations of the present invention include the following preferred combinations:

S No.	I	II	III	IV
236	Mancozeb	Bixafen	Cyproconazole	—
237	Mancozeb	Bixafen	Difenoconazole	—
238	Mancozeb	Bixafen	Epoxiconazole	—
239	Mancozeb	Bixafen	Hexaconazole	—
240	Mancozeb	Bixafen	Tebuconazole	—
241	Mancozeb	Bixafen	Tetraconazole	—
242	Mancozeb	Bixafen	Prothioconazole	—
243	Mancozeb	Bixafen	—	Azoxystrobin
244	Mancozeb	Bixafen	—	Picoxystrobin
245	Mancozeb	Bixafen	—	Pyraclostrobin
246	Mancozeb	Bixafen	—	Kresoxim-methyl
247	Mancozeb	Bixafen	—	Trifloxystrobin
248	Mancozeb	Bixafen	Cyproconazole	Azoxystrobin
249	Mancozeb	Bixafen	Cyproconazole	Picoxystrobin
250	Mancozeb	Bixafen	Cyproconazole	Pyraclostrobin
251	Mancozeb	Bixafen	Cyproconazole	Kresoxim-methyl
252	Mancozeb	Bixafen	Cyproconazole	Trifloxystrobin
253	Mancozeb	Bixafen	Difenoconazole	Azoxystrobin
254	Mancozeb	Bixafen	Difenoconazole	Picoxystrobin
255	Mancozeb	Bixafen	Difenoconazole	Pyraclostrobin
256	Mancozeb	Bixafen	Difenoconazole	Kresoxim-methyl
257	Mancozeb	Bixafen	Difenoconazole	Trifloxystrobin
258	Mancozeb	Bixafen	Epoxiconazole	Azoxystrobin
259	Mancozeb	Bixafen	Epoxiconazole	Picoxystrobin
260	Mancozeb	Bixafen	Epoxiconazole	Pyraclostrobin
261	Mancozeb	Bixafen	Epoxiconazole	Kresoxim-methyl
262	Mancozeb	Bixafen	Epoxiconazole	Trifloxystrobin
263	Mancozeb	Bixafen	Hexaconazole	Azoxystrobin
264	Mancozeb	Bixafen	Hexaconazole	Picoxystrobin
265	Mancozeb	Bixafen	Hexaconazole	Pyraclostrobin
266	Mancozeb	Bixafen	Hexaconazole	Kresoxim-methyl

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S No.	I	II	III	IV
267	Mancozeb	Bixafen	Hexaconazole	Trifloxystrobin
268	Mancozeb	Bixafen	Tebuconazole	Azoxystrobin
269	Mancozeb	Bixafen	Tebuconazole	Picoxystrobin
270	Mancozeb	Bixafen	Tebuconazole	Pyraclostrobin
271	Mancozeb	Bixafen	Tebuconazole	Kresoxim-methyl
272	Mancozeb	Bixafen	Tebuconazole	Trifloxystrobin
273	Mancozeb	Bixafen	Tetraconazole	Azoxystrobin
274	Mancozeb	Bixafen	Tetraconazole	Picoxystrobin
275	Mancozeb	Bixafen	Tetraconazole	Pyraclostrobin
276	Mancozeb	Bixafen	Tetraconazole	Kresoxim-methyl
277	Mancozeb	Bixafen	Tetraconazole	Trifloxystrobin
278	Mancozeb	Bixafen	Prothioconazole	Azoxystrobin
279	Mancozeb	Bixafen	Prothioconazole	Picoxystrobin
280	Mancozeb	Bixafen	Prothioconazole	Pyraclostrobin
281	Mancozeb	Bixafen	Prothioconazole	Kresoxim-methyl
282	Mancozeb	Bixafen	Prothioconazole	Trifloxystrobin

In an embodiment of the combinations of the present invention, the preferred succinate dehydrogenase inhibitor fungicide is Fluxapyroxad.

In an embodiment, the combinations of the present invention include the following preferred combinations:

S No.	I	II	III	IV
283	Mancozeb	Fluxapyroxad	Cyproconazole	—
284	Mancozeb	Fluxapyroxad	Difenoconazole	—
285	Mancozeb	Fluxapyroxad	Epoxiconazole	—
286	Mancozeb	Fluxapyroxad	Hexaconazole	—
287	Mancozeb	Fluxapyroxad	Tebuconazole	—
288	Mancozeb	Fluxapyroxad	Tetraconazole	—
289	Mancozeb	Fluxapyroxad	Prothioconazole	—
290	Mancozeb	Fluxapyroxad	—	Azoxystrobin
291	Mancozeb	Fluxapyroxad	—	Picoxystrobin
292	Mancozeb	Fluxapyroxad	—	Pyraclostrobin
293	Mancozeb	Fluxapyroxad	—	Kresoxim-methyl
294	Mancozeb	Fluxapyroxad	—	Trifloxystrobin
295	Mancozeb	Fluxapyroxad	Cyproconazole	Azoxystrobin
296	Mancozeb	Fluxapyroxad	Cyproconazole	Picoxystrobin
297	Mancozeb	Fluxapyroxad	Cyproconazole	Pyraclostrobin
298	Mancozeb	Fluxapyroxad	Cyproconazole	Kresoxim-methyl
299	Mancozeb	Fluxapyroxad	Cyproconazole	Trifloxystrobin
300	Mancozeb	Fluxapyroxad	Difenoconazole	Azoxystrobin
301	Mancozeb	Fluxapyroxad	Difenoconazole	Picoxystrobin
302	Mancozeb	Fluxapyroxad	Difenoconazole	Pyraclostrobin
303	Mancozeb	Fluxapyroxad	Difenoconazole	Kresoxim-methyl
304	Mancozeb	Fluxapyroxad	Difenoconazole	Trifloxystrobin
305	Mancozeb	Fluxapyroxad	Epoxiconazole	Azoxystrobin
306	Mancozeb	Fluxapyroxad	Epoxiconazole	Picoxystrobin
307	Mancozeb	Fluxapyroxad	Epoxiconazole	Pyraclostrobin
308	Mancozeb	Fluxapyroxad	Epoxiconazole	Kresoxim-methyl
309	Mancozeb	Fluxapyroxad	Epoxiconazole	Trifloxystrobin
310	Mancozeb	Fluxapyroxad	Hexaconazole	Azoxystrobin
311	Mancozeb	Fluxapyroxad	Hexaconazole	Picoxystrobin
312	Mancozeb	Fluxapyroxad	Hexaconazole	Pyraclostrobin
313	Mancozeb	Fluxapyroxad	Hexaconazole	Kresoxim-methyl
314	Mancozeb	Fluxapyroxad	Hexaconazole	Trifloxystrobin
315	Mancozeb	Fluxapyroxad	Tebuconazole	Azoxystrobin
316	Mancozeb	Fluxapyroxad	Tebuconazole	Picoxystrobin
317	Mancozeb	Fluxapyroxad	Tebuconazole	Pyraclostrobin
318	Mancozeb	Fluxapyroxad	Tebuconazole	Kresoxim-methyl
319	Mancozeb	Fluxapyroxad	Tebuconazole	Trifloxystrobin
320	Mancozeb	Fluxapyroxad	Tetraconazole	Azoxystrobin
321	Mancozeb	Fluxapyroxad	Tetraconazole	Picoxystrobin
322	Mancozeb	Fluxapyroxad	Tetraconazole	Pyraclostrobin

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S No.	I	II	III	IV
323	Mancozeb	Fluxapyroxad	Tetraconazole	Kresoxim-methyl
324	Mancozeb	Fluxapyroxad	Tetraconazole	Trifloxystrobin
325	Mancozeb	Fluxapyroxad	Prothioconazole	Azoxystrobin
326	Mancozeb	Fluxapyroxad	Prothioconazole	Picoxystrobin
327	Mancozeb	Fluxapyroxad	Prothioconazole	Pyraclostrobin
328	Mancozeb	Fluxapyroxad	Prothioconazole	Kresoxim-methyl
329	Mancozeb	Fluxapyroxad	Prothioconazole	Trifloxystrobin

In an embodiment of the combinations of the present invention, the preferred succinate dehydrogenase inhibitor fungicide is Furametpyr.

In an embodiment, the combinations of the present invention include the following preferred combinations:

S No.	I	II	III	IV
330	Mancozeb	Furametpyr	Cyproconazole	—
331	Mancozeb	Furametpyr	Difenoconazole	—
332	Mancozeb	Furametpyr	Epoxiconazole	—
333	Mancozeb	Furametpyr	Hexaconazole	—
334	Mancozeb	Furametpyr	Tebuconazole	—
335	Mancozeb	Furametpyr	Tetraconazole	—
336	Mancozeb	Furametpyr	Prothioconazole	—
337	Mancozeb	Furametpyr	—	Azoxystrobin
338	Mancozeb	Furametpyr	—	Picoxystrobin
339	Mancozeb	Furametpyr	—	Pyraclostrobin
340	Mancozeb	Furametpyr	—	Kresoxim-methyl
341	Mancozeb	Furametpyr	—	Trifloxystrobin
342	Mancozeb	Furametpyr	Cyproconazole	Azoxystrobin
343	Mancozeb	Furametpyr	Cyproconazole	Picoxystrobin
344	Mancozeb	Furametpyr	Cyproconazole	Pyraclostrobin
345	Mancozeb	Furametpyr	Cyproconazole	Kresoxim-methyl
346	Mancozeb	Furametpyr	Cyproconazole	Trifloxystrobin
347	Mancozeb	Furametpyr	Difenoconazole	Azoxystrobin
348	Mancozeb	Furametpyr	Difenoconazole	Picoxystrobin
349	Mancozeb	Furametpyr	Difenoconazole	Pyraclostrobin
350	Mancozeb	Furametpyr	Difenoconazole	Kresoxim-methyl
351	Mancozeb	Furametpyr	Difenoconazole	Trifloxystrobin
352	Mancozeb	Furametpyr	Epoxiconazole	Azoxystrobin
353	Mancozeb	Furametpyr	Epoxiconazole	Picoxystrobin
354	Mancozeb	Furametpyr	Epoxiconazole	Pyraclostrobin
355	Mancozeb	Furametpyr	Epoxiconazole	Kresoxim-methyl
356	Mancozeb	Furametpyr	Epoxiconazole	Trifloxystrobin
357	Mancozeb	Furametpyr	Hexaconazole	Azoxystrobin
358	Mancozeb	Furametpyr	Hexaconazole	Picoxystrobin
359	Mancozeb	Furametpyr	Hexaconazole	Pyraclostrobin
360	Mancozeb	Furametpyr	Hexaconazole	Kresoxim-methyl
361	Mancozeb	Furametpyr	Hexaconazole	Trifloxystrobin
362	Mancozeb	Furametpyr	Tebuconazole	Azoxystrobin
363	Mancozeb	Furametpyr	Tebuconazole	Picoxystrobin
364	Mancozeb	Furametpyr	Tebuconazole	Pyraclostrobin
365	Mancozeb	Furametpyr	Tebuconazole	Kresoxim-methyl
366	Mancozeb	Furametpyr	Tebuconazole	Trifloxystrobin
367	Mancozeb	Furametpyr	Tetraconazole	Azoxystrobin
368	Mancozeb	Furametpyr	Tetraconazole	Picoxystrobin
369	Mancozeb	Furametpyr	Tetraconazole	Pyraclostrobin
370	Mancozeb	Furametpyr	Tetraconazole	Kresoxim-methyl
371	Mancozeb	Furametpyr	Tetraconazole	Trifloxystrobin
372	Mancozeb	Furametpyr	Prothioconazole	Azoxystrobin
373	Mancozeb	Furametpyr	Prothioconazole	Picoxystrobin
374	Mancozeb	Furametpyr	Prothioconazole	Pyraclostrobin
375	Mancozeb	Furametpyr	Prothioconazole	Kresoxim-methyl
376	Mancozeb	Furametpyr	Prothioconazole	Trifloxystrobin

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In an embodiment of the combinations of the present invention, the preferred succinate dehydrogenase inhibitor fungicide is Penflufen.

In an embodiment, the combinations of the present invention include the following preferred combinations:

S No.	I	II	III	IV
377	Mancozeb	Penflufen	Cyproconazole	—
378	Mancozeb	Penflufen	Difenoconazole	—
379	Mancozeb	Penflufen	Epoxiconazole	—
380	Mancozeb	Penflufen	Hexaconazole	—
381	Mancozeb	Penflufen	Tebuconazole	—
382	Mancozeb	Penflufen	Tetraconazole	—
383	Mancozeb	Penflufen	Prothioconazole	—
384	Mancozeb	Penflufen	—	Azoxystrobin
385	Mancozeb	Penflufen	—	Picoxystrobin
386	Mancozeb	Penflufen	—	Pyraclostrobin
387	Mancozeb	Penflufen	—	Kresoxim-methyl
388	Mancozeb	Penflufen	—	Trifloxystrobin
389	Mancozeb	Penflufen	Cyproconazole	Azoxystrobin
390	Mancozeb	Penflufen	Cyproconazole	Picoxystrobin
391	Mancozeb	Penflufen	Cyproconazole	Pyraclostrobin
392	Mancozeb	Penflufen	Cyproconazole	Kresoxim-methyl
393	Mancozeb	Penflufen	Cyproconazole	Trifloxystrobin
394	Mancozeb	Penflufen	Difenoconazole	Azoxystrobin
395	Mancozeb	Penflufen	Difenoconazole	Picoxystrobin
396	Mancozeb	Penflufen	Difenoconazole	Pyraclostrobin
397	Mancozeb	Penflufen	Difenoconazole	Kresoxim-methyl
398	Mancozeb	Penflufen	Difenoconazole	Trifloxystrobin
399	Mancozeb	Penflufen	Epoxiconazole	Azoxystrobin
400	Mancozeb	Penflufen	Epoxiconazole	Picoxystrobin
401	Mancozeb	Penflufen	Epoxiconazole	Pyraclostrobin
402	Mancozeb	Penflufen	Epoxiconazole	Kresoxim-methyl
403	Mancozeb	Penflufen	Epoxiconazole	Trifloxystrobin
404	Mancozeb	Penflufen	Hexaconazole	Azoxystrobin
405	Mancozeb	Penflufen	Hexaconazole	Picoxystrobin
406	Mancozeb	Penflufen	Hexaconazole	Pyraclostrobin
407	Mancozeb	Penflufen	Hexaconazole	Kresoxim-methyl
408	Mancozeb	Penflufen	Hexaconazole	Trifloxystrobin
409	Mancozeb	Penflufen	Tebuconazole	Azoxystrobin
410	Mancozeb	Penflufen	Tebuconazole	Picoxystrobin
411	Mancozeb	Penflufen	Tebuconazole	Pyraclostrobin
412	Mancozeb	Penflufen	Tebuconazole	Kresoxim-methyl
413	Mancozeb	Penflufen	Tebuconazole	Trifloxystrobin
414	Mancozeb	Penflufen	Tetraconazole	Azoxystrobin
415	Mancozeb	Penflufen	Tetraconazole	Picoxystrobin
416	Mancozeb	Penflufen	Tetraconazole	Pyraclostrobin
417	Mancozeb	Penflufen	Tetraconazole	Kresoxim-methyl
418	Mancozeb	Penflufen	Tetraconazole	Trifloxystrobin
419	Mancozeb	Penflufen	Prothioconazole	Azoxystrobin
420	Mancozeb	Penflufen	Prothioconazole	Picoxystrobin
421	Mancozeb	Penflufen	Prothioconazole	Pyraclostrobin
422	Mancozeb	Penflufen	Prothioconazole	Kresoxim-methyl
423	Mancozeb	Penflufen	Prothioconazole	Trifloxystrobin

In an embodiment of the combinations of the present invention, the preferred succinate dehydrogenase inhibitor fungicide is Sedaxane.

In an embodiment, the combinations of the present invention include the following preferred combinations:

S No.	I	II	III	IV
424	Mancozeb	Sedaxane	Cyproconazole	—
425	Mancozeb	Sedaxane	Difenoconazole	—
426	Mancozeb	Sedaxane	Epoxiconazole	—
427	Mancozeb	Sedaxane	Hexaconazole	—

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-continued

S No.	I	II	III	IV
428	Mancozeb	Sedaxane	Tebuconazole	—
429	Mancozeb	Sedaxane	Tetraconazole	—
430	Mancozeb	Sedaxane	Prothioconazole	—
431	Mancozeb	Sedaxane	—	Azoxystrobin
432	Mancozeb	Sedaxane	—	Picoxystrobin
433	Mancozeb	Sedaxane	—	Pyraclostrobin
434	Mancozeb	Sedaxane	—	Kresoxim-methyl
435	Mancozeb	Sedaxane	—	Trifloxystrobin
436	Mancozeb	Sedaxane	Cyproconazole	Azoxystrobin
437	Mancozeb	Sedaxane	Cyproconazole	Picoxystrobin
438	Mancozeb	Sedaxane	Cyproconazole	Pyraclostrobin
439	Mancozeb	Sedaxane	Cyproconazole	Kresoxim-methyl
440	Mancozeb	Sedaxane	Cyproconazole	Trifloxystrobin
441	Mancozeb	Sedaxane	Difenoconazole	Azoxystrobin
442	Mancozeb	Sedaxane	Difenoconazole	Picoxystrobin
443	Mancozeb	Sedaxane	Difenoconazole	Pyraclostrobin
444	Mancozeb	Sedaxane	Difenoconazole	Kresoxim-methyl
445	Mancozeb	Sedaxane	Difenoconazole	Trifloxystrobin
446	Mancozeb	Sedaxane	Epoxiconazole	Azoxystrobin
447	Mancozeb	Sedaxane	Epoxiconazole	Picoxystrobin
448	Mancozeb	Sedaxane	Epoxiconazole	Pyraclostrobin
449	Mancozeb	Sedaxane	Epoxiconazole	Kresoxim-methyl
450	Mancozeb	Sedaxane	Epoxiconazole	Trifloxystrobin
451	Mancozeb	Sedaxane	Hexaconazole	Azoxystrobin
452	Mancozeb	Sedaxane	Hexaconazole	Picoxystrobin
453	Mancozeb	Sedaxane	Hexaconazole	Pyraclostrobin
454	Mancozeb	Sedaxane	Hexaconazole	Kresoxim-methyl
455	Mancozeb	Sedaxane	Hexaconazole	Trifloxystrobin
456	Mancozeb	Sedaxane	Tebuconazole	Azoxystrobin
457	Mancozeb	Sedaxane	Tebuconazole	Picoxystrobin
458	Mancozeb	Sedaxane	Tebuconazole	Pyraclostrobin
459	Mancozeb	Sedaxane	Tebuconazole	Kresoxim-methyl
460	Mancozeb	Sedaxane	Tebuconazole	Trifloxystrobin
461	Mancozeb	Sedaxane	Tetraconazole	Azoxystrobin
462	Mancozeb	Sedaxane	Tetraconazole	Picoxystrobin
463	Mancozeb	Sedaxane	Tetraconazole	Pyraclostrobin
464	Mancozeb	Sedaxane	Tetraconazole	Kresoxim-methyl
465	Mancozeb	Sedaxane	Tetraconazole	Trifloxystrobin
466	Mancozeb	Sedaxane	Prothioconazole	Azoxystrobin
467	Mancozeb	Sedaxane	Prothioconazole	Picoxystrobin
468	Mancozeb	Sedaxane	Prothioconazole	Pyraclostrobin
469	Mancozeb	Sedaxane	Prothioconazole	Kresoxim-methyl
470	Mancozeb	Sedaxane	Prothioconazole	Trifloxystrobin

The combinations of the present invention may be formulated in the form of a composition.

In an embodiment, the present invention may provide a composition comprising:

- (a) at least one succinate dehydrogenase inhibitor fungicide;
- (b) at least one dithiocarbamate fungicide;
- (c) at least one quinone outside inhibitor; and
- (d) at least one agrochemically acceptable excipient.

In an embodiment, the present invention may provide a composition comprising:

- (a) at least one succinate dehydrogenase inhibitor fungicide;
- (b) at least one dithiocarbamate fungicide;
- (c) at least one ergosterol biosynthesis inhibitor; and
- (d) at least one agrochemically acceptable excipient.

In an embodiment, the present invention may provide a composition comprising:

- (a) at least one succinate dehydrogenase inhibitor fungicide;
- (b) at least one dithiocarbamate fungicide;

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- (c) at least one a quinone outside inhibitor;
- (d) at least one ergosterol biosynthesis inhibitor; and
- (e) at least one agrochemically acceptable excipient.

The amount of a composition according to the invention to be applied, will depend on various factors, such as the subject of the treatment, such as, for example plants, soil or seeds; the type of treatment, such as, for example spraying, dusting or seed dressing; the purpose of the treatment, such as, for example prophylactic or therapeutic disease control; in case of disease control the type of fungi to be controlled or the application time. This amount of the combinations of the present invention to be applied can be readily deduced by a skilled agronomist.

Thus in an embodiment, the present invention may provide compositions comprising:

- (a) at least one pyrazaole carboxamide selected from benzovindiflupyr, bixafen, fluxapyroxad, furametpyr, isopyrazam, penflufen, penthiopyrad, and sedaxane;
 - (b) at least one quinone outside inhibitor, and/or at least one ergosterol biosynthesis inhibitor; and
 - (c) at least one dithiocarbamate fungicide;
- said fungicides being combined in agrochemically acceptable amounts.

In an embodiment, the total amount of succinate dehydrogenase inhibitor in the composition may typically be in the range of 0.1 to 99% by weight, preferably 0.2 to 90% by weight. The total amount of dithiocarbamate fungicide in the composition may be in the range of 0.1 to 99% by weight. The total amount of ergosterol biosynthesis inhibitor in the composition may be in the range of 0.1 to 99% by weight. The total amount of Quinone outside inhibitor in the composition may be in the range of 0.1 to 99% by weight.

In an embodiment, the constituent fungicides of the combination of the present invention may be admixed in ratio of (1-80):(1-80):(1-80) of the dithiocarbamate fungicide, succinate dehydrogenase inhibitor fungicide and the third fungicide respectively.

In an embodiment, the constituents of the composition of the present invention may be tank mixed and sprayed at the locus of the infection, or may be alternatively be mixed with surfactants and then sprayed.

In an embodiment, the constituents of the composition of the present invention may be used for foliar application, ground or applications to plant propagation materials.

In an embodiment, the compositions of the present invention may typically be produce by mixing the actives in the composition with an inert carrier, and adding surfactants and other adjuvants and carriers as needed and formulated into solid, or liquid formulations, including but not limited to wettable powders, granules, dusts, Soluble (liquid) concentrates, suspension concentrates, oil in water emulsion, water in oil emulsion, emulsifiable concentrates, capsule suspensions, ZC formulations, oil dispersions or other known formulation types. The composition may also be used for treatment of a plant propagation material such as seeds etc.

Examples of the solid carrier used in formulation include fine powders or granules such as minerals such as kaolin clay, attapulgite clay, bentonite, montmorillonite, acid white clay, pyrophyllite, talc, diatomaceous earth and calcite; natural organic materials such as corn rachis powder and walnut husk powder; synthetic organic materials such as urea; salts such as calcium carbonate and ammonium sulfate; synthetic inorganic materials such as synthetic hydrated silicon oxide; and as a liquid carrier, aromatic hydrocarbons such as xylene, alkylbenzene and methylnaphthalene; alcohols such as 2-propanol, ethyleneglycol, propylene glycol, and ethylene glycol monoethyl ether; ketones such as

acetone, cyclohexanone and isophorone; vegetable oil such as soybean oil and cotton seed oil; petroleum aliphatic hydrocarbons, esters, dimethylsulfoxide, acetonitrile and water.

Examples of the surfactant include anionic surfactants such as alkyl sulfate ester salts, alkylaryl sulfonate salts, dialkyl sulfosuccinate salts, polyoxyethylene alkylaryl ether phosphate ester salts, lignosulfonate salts and naphthalene sulfonate formaldehyde polycondensates; and nonionic surfactants such as polyoxyethylene alkyl aryl ethers, polyoxyethylene alkylpolyoxypropylene block copolymers and sorbitan fatty acid esters and cationic surfactants such as alkyltrimethylammonium salts.

Examples of the other formulation auxiliary agents include water-soluble polymers such as polyvinyl alcohol and polyvinylpyrrolidone, polysaccharides such as Arabic gum, alginic acid and the salt thereof, CMC (carboxymethyl-cellulose), Xanthan gum, inorganic materials such as aluminum magnesium silicate and alumina sol, preservatives, coloring agents and stabilization agents such as PAP (acid phosphate isopropyl) and BHT.

The compositions according to the present invention is effective for the following plant diseases:

Disease in rice: Blast (*Magnaporthe grisea*), Helminthosporium leaf spot (*Cochliobolus miyabeanus*), sheath blight (*Rhizoctonia solani*), and bakanae disease (*Gibberella fujikuroi*).

Diseases in wheat: powdery mildew (*Erysiphe graminis*), Fusarium head blight (*Fusarium graminearum*, *F. avenacerum*, *F. culmorum*, *Microdochium nivale*), rust (*Puccinia striiformis*, *P. graminis*, *P. recondita*), pink snow mold (*Micronectriella nivale*), *Typhula* snow blight (*Typhula* sp.), loose smut (*Ustilago tritici*), bunt (*Tilletia caries*), eyespot (*Pseudocercospora herpotrichoides*), leaf blotch (*Mycosphaerella graminicola*), glume blotch (*Stagonospora nodorum*), septoria, and yellow spot (*Pyrenophora tritici-repentis*).

Diseases of barley: powdery mildew (*Erysiphe graminis*), *Fusarium* head blight (*Fusarium graminearum*, *F. avenacerum*, *F. culmorum*, *Microdochium nivale*), rust (*Puccinia striiformis*, *P. graminis*, *P. hordei*), loose smut (*Ustilago nuda*), scald (*Rhynchosporium secalis*), net blotch (*Pyrenophora teres*), spot blotch (*Cochliobolus sativus*), leaf stripe (*Pyrenophora graminea*), and *Rhizoctonia* damping-off (*Rhizoctonia solani*).

Diseases in corn: smut (*Ustilago maydis*), brown spot (*Cochliobolus heterostrophus*), copper spot (*Gloeocercospora sorghi*), southern rust (*Puccinia polysora*), gray leaf spot (*Cercospora zeae-maydis*), white spot (*Phaeosphaeria mydis* and/or *Pantoea ananatis*) and *Rhizoctonia* damping-off (*Rhizoctonia solani*).

Diseases of citrus: melanose (*Diaporthe citri*), scab (*Elsinoe fawcetti*), penicillium rot (*Penicillium digitatum*, *P. italicum*), and brown rot (*Phytophthora parasitica*, *Phytophthora citrophthora*).

Diseases of apple: blossom blight (*Monilinia mali*), canker (*Valsa ceratosperma*), powdery mildew (*Podosphaera leucotricha*), *Alternaria* leaf spot (*Alternaria alternata* apple pathotype), scab (*Venturia inaequalis*), powdery mildew, bitter rot (*Colletotrichum acutatum*), crown rot (*Phytophthora cactorum*), blotch (*Diplocarpon mali*), and ring rot (*Botryosphaeria berengeriana*).

Diseases of pear: scab (*Venturia nashicola*, *V. pirina*), powdery mildew, black spot (*Alternaria alternata* Japa-

nese pear pathotype), rust (*Gymnosporangium haraeae-num*), and *phytophthora* fruit rot (*Phytophthora cactorum*).

Diseases of peach: brown rot (*Monilinia fructicola*), powdery mildew, scab (*Cladosporium carpophilum*), and *phomopsis* rot (*Phomopsis* sp.).

Diseases of grape: anthracnose (*Elsinoe ampelina*), ripe rot (*Glomerella cingulata*), powdery mildew (*Uncinula necator*), rust (*Phakopsora ampelopsidis*), black rot (*Guignardia bidwellii*), *botrytis*, and downy mildew (*Plasmopara viticola*).

Diseases of Japanese persimmon: anthracnose (*Gloeosporium kaki*), and leaf spot (*Cercospora kaki*, *Mycosphaerella nawae*).

Diseases of gourd: anthracnose (*Colletotrichum lagenarium*), powdery mildew (*Sphaerotheca fuliginea*), gummy stem blight (*Mycosphaerella melonis*), *Fusarium* wilt (*Fusarium oxysporum*), downy mildew (*Pseudoperonospora cubensis*), *Phytophthora* rot (*Phytophthora* sp.), and damping-off (*Pythium* sp.).

Diseases of tomato: early blight (*Alternaria solani*), leaf mold (*Cladosporium fulvum*), and late blight (*Phytophthora infestans*).

Diseases of eggplant: brown spot (*Phomopsis vexans*), and powdery mildew (*Erysiphe cichoracearum*).

Diseases of cruciferous vegetables: *Alternaria* leaf spot (*Alternaria japonica*), white spot (*Cercospora brassicae*), clubroot (*Plasmodiophora brassicae*), and downy mildew (*Peronospora parasitica*).

Diseases of onion: rust (*Puccinia allii*), and downy mildew (*Peronospora destructor*).

Diseases of soybean: purple seed stain (*Cercospora kikuchii*), sphaceloma scab (*Elsinoe glycines*), pod and stem blight (*Diaporthe phaseolorum* var. *sojae*), *septoria* brown spot (*Septoria glycines*), frog-eye leaf spot (*Cercospora sojae*), rust (*Phakopsora pachyrhizi*), Yellow rust, brown stem rot (*Phytophthora sojae*), and *Rhizoctonia* damping-off (*Rhizoctonia solani*).

Diseases of kidney bean: anthracnose (*Colletotrichum lindemthianum*).

Diseases of peanut: leaf spot (*Cercospora personata*), brown leaf spot (*Cercospora arachidicola*) and southern blight (*Sclerotium rolfsii*).

Diseases of garden pea: powdery mildew (*Erysiphe pisi*), and root rot (*Fusarium solani* f. sp. *pisi*).

Diseases of potato: early blight (*Alternaria solani*), late blight (*Phytophthora infestans*), pink rot (*Phytophthora erythroseptica*), and powdery scab (*Spongospora subterranean* f. sp. *subterranea*).

Diseases of strawberry: powdery mildew (*Sphaerotheca humuli*), and anthracnose (*Glomerella cingulata*).

Diseases of tea: net blister blight (*Exobasidium reticulatum*), white scab (*Elsinoe leucospila*), gray blight (*Pestalotiopsis* sp.), and anthracnose (*Colletotrichum theae-sinensis*).

Diseases of tobacco: brown spot (*Alternaria longipes*), powdery mildew (*Erysiphe cichoracearum*), anthracnose (*Colletotrichum tabacum*), downy mildew (*Peronospora tabacina*), and black shank (*Phytophthora nicotianae*).

Diseases of rapeseed: sclerotinia rot (*Sclerotinia sclerotiorum*), and *Rhizoctonia* damping-off (*Rhizoctonia solani*).

Diseases of cotton: *Rhizoctonia* damping-off (*Rhizoctonia solani*).

Diseases of sugar beat: *Cercospora* leaf spot (*Cercospora beticola*), leaf blight (*Thanatephorus cucumeris*), Root

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rot (*Thanatephorus cucumeris*), and *Aphanomyces* root rot (*Aphanomyces cochlioides*).

Diseases of rose: black spot (*Diplocarpon rosae*), powdery mildew (*Sphaerotheca pannosa*), and downy mildew (*Peronospora sparsa*).

Diseases of chrysanthemum and asteraceous plants: downy mildew (*Bremia lactucae*), leaf blight (*Septoria chrysanthemi-indici*), and white rust (*Puccinia horiana*).

Diseases of various groups: diseases caused by *Pythium* spp. (*Pythium aphanidermatum*, *Pythium debarianum*, *Pythium graminicola*, *Pythium irregulare*, *Pythium ultimum*), gray mold (*Botrytis cinerea*), and *Sclerotinia* rot (*Sclerotinia sclerotiorum*).

Disease of Japanese radish: *Alternaria* leaf spot (*Alternaria brassicicola*).

Diseases of turfgrass: dollar spot (*Sclerotinia homeocarpa*), and brown patch and large patch (*Rhizoctonia solani*).

Disease of banana: Black sigatoka (*Mycosphaerella fijiensis*), Yellow sigatoka (*Mycosphaerella musicola*).

Disease of sunflower: downy mildew (*Plasmopara halstedii*).

Seed diseases or diseases in the early stages of the growth of various plants caused by *Aspergillus* spp., *Penicillium* spp., *Fusarium* spp., *Gibberella* spp., *Tricoderma* spp., *Thielaviopsis* spp., *Rhizopus* spp., *Mucor* spp., *Corticium* spp., *Phoma* spp., *Rhizoctonia* spp. and *Diplodia* spp.

Viral diseases of various plants mediated by *Polymixa* spp. or *Olpidium* spp. and so on.

The compositions of the present invention can be used in agricultural lands such as fields, paddy fields, lawns and orchards or in non-agricultural lands. The present invention may be used to control diseases in agricultural lands for cultivating the plants without any phytotoxicity to the plant.

Examples of the crops on which the present compositions may be used include but are not limited to corn, rice, wheat, barley, rye, oat, sorghum, cotton, soybean, peanut, buckwheat, beet, rapeseed, sunflower, sugar cane, tobacco, etc.; vegetables: solanaceous vegetables such as eggplant, tomato, pimento, pepper, potato, etc., cucurbit vegetables such as cucumber, pumpkin, zucchini, water melon, melon, squash, etc., cruciferous vegetables such as radish, white turnip, horseradish, kohlrabi, Chinese cabbage, cabbage, leaf mustard, broccoli, cauliflower, etc., asteraceous vegetables such as burdock, crown daisy, artichoke, lettuce, etc., liliaceous vegetables such as green onion, onion, garlic, and asparagus, ammiaceous vegetables such as carrot, parsley, celery, parsnip, etc., chenopodiaceous vegetables such as spinach, Swiss chard, etc., lamiaceous vegetables such as *Perilla frutescens*, mint, basil, etc., strawberry, sweet potato, *Dioscorea japonica*, colocasia, etc., flowers, foliage plants, turf grasses, fruits: pome fruits such as apple, pear, quince, etc., stone fleshy fruits such as peach, plum, nectarine, *Prunus mume*, cherry fruit, apricot, prune, etc., citrus fruits such as orange, lemon, lime, grapefruit, etc., nuts such as chestnuts, walnuts, hazelnuts, almond, pistachio, cashew nuts, macadamia nuts, etc. berries such as blueberry, cranberry, blackberry, raspberry, etc., grape, kaki fruit, olive, plum, banana, coffee, date palm, coconuts, etc., trees other than fruit trees; tea, mulberry, flowering plant, trees such as ash, birch, dogwood, Eucalyptus, Ginkgo biloba, lilac, maple, Quercus, poplar, Judas tree, Liquidambar formosana, plane tree, zelkova, Japanese arborvitae, fir wood, hemlock, juniper, Pinus, Picea, and Taxus cuspidate, etc.

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In an embodiment, the constituent fungicides of the combination of the present invention may be admixed in ratio of (1-80):(1-80):(1-80):(1-80)

In an aspect, the present invention may provide methods of controlling fungal diseases comprising applying a combination comprising:

(a) at least one succinate dehydrogenase inhibitor fungicide;

(b) at least one quinone outside inhibitor and/or at least one ergosterol biosynthesis inhibitor; and

(c) at least one dithiocarbamate fungicide.

In an embodiment, the succinate dehydrogenase inhibitor fungicide, the quinone outside inhibitor fungicide, the ergosterol biosynthesis inhibitor fungicide, and the dithiocarbamate fungicide may be selected according to any of the preferred embodiments of the combinations described hereinabove.

The combinations of the present invention may be sold as a pre-mix composition or a kit of parts such that individual actives may be mixed before spraying. Alternatively, the kit of parts may contain succinate dehydrogenase inhibitor fungicide and the dithiocarbamate fungicide pre-mixed and the third active may be admixed with an adjuvant such that the two components may be tank mixed before spraying.

The composition of the present invention may be applied simultaneously as a tank mix or a formulation or may be applied sequentially. The application may be made to the soil before emergence of the plants, either pre-planting or post-planting. The application may be made as a foliar spray at different timings during crop development, with either one or two applications early or late post-emergence.

The compositions according to the invention can be applied before or after infection of the useful plants or the propagation material thereof by the fungi.

As will be demonstrated in the examples, the addition of a dithiocarbamate fungicide to a combination of succinate dehydrogenase inhibitors which are combined with Quinone outside inhibitors and/or ergosterol biosynthesis inhibitors, greatly improved the disease control as well as improved yield and demonstrated a synergistic effect. The lower the mixture performance in the disease control, the greater the additional benefit of the mancozeb when added to the compositions of the present invention.

Examples: Studies were conducted to study the addition of dithiocarbamate fungicides to succinate dehydrogenase inhibitor fungicides and at least one other fungicide and the contribution of the dithiocarbamate to the efficacy of these mixtures. Experiments were conducted over a period of two years to study the effect of the addition of dithiocarbamates on the efficacy of succinate dehydrogenase inhibitors alone and when combined with a co-fungicide such as a Quinone outside inhibitors and/or ergosterol biosynthesis inhibitor fungicide. Doses tested were at rates of 1500 g/ha for Mancozeb, 150 ml/ha for the ergosterol biosynthesis inhibitor, 200 g/ha for the Quinone outside inhibitor, 1000 g/ha and 200 g/ha for succinate dehydrogenase inhibitor fungicides. The combinations were tested for efficacy of disease control in soybean for the control of Asian soybean rust. The trials were carried out at various locations in India.

The tests were conducted on soybean cultivar Monsoy 9144 RR. The commercially available individual active ingredients were used at the indicated dosages.

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TABLE 1

Table 1 shows the efficacy of mancozeb when added to combinations containing succinate dehydrogenase inhibitor fungicides and ergosterol biosynthesis inhibitor.			
Treatment	Dose rates (ml/g/ha)	Mean percent disease control	
		2015/16	2016/17
Prothioconazole + Benzovindiflupyr	150 + 200	88.04	71.04
Prothioconazole + Benzovindiflupyr + Mancozeb	150 + 200 + 1500	95.07	95.07
Prothioconazole + Isopyrazam	150 + 1000	87.14	70.23
Prothioconazole + Isopyrazam + Mancozeb	150 + 1000 + 1500	93.4	93.07
Prothioconazole + Penthiopyrad	150 + 1000	83.33	68.15
Prothioconazole + Penthiopyrad + Mancozeb	150 + 1000 + 1500	93.07	92.18

Table 1 clearly demonstrates the increase in control when mancozeb is added to the combination of ergosterol biosynthesis inhibitors and succinate dehydrogenase inhibitor fungicides.

TABLE 2

Table 2 demonstrates efficacy when mancozeb is added to combinations of succinate dehydrogenase inhibitor fungicides and quinone outside inhibitor fungicides and ergosterol biosynthesis inhibitors.			
Treatment	Dose rates (ml/g/ha)	Mean percent disease control	
		2015/16	2016/17
Prothioconazole + Benzovindiflupyr + Azoxystrobin	150 + 200 + 500	93.07	91.07
Prothioconazole + Benzovindiflupyr + Azoxystrobin + Mancozeb	150 + 200 + 500 + 1500	96.36	95.03

Table 2 clearly demonstrates the importance of adding mancozeb to the combination. The addition of mancozeb improved disease control.

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It was thus found that the incorporation of mancozeb greatly increased efficacy and disease control of the succinate dehydrogenase inhibitor fungicide treatments when mixed with ergosterol biosynthesis inhibitors or quinone outside inhibitors. It was thus concluded that the addition of mancozeb enhanced the efficacy of the combination and surprisingly gave a synergistic effect. The addition of a dithiocarbamate increased disease control and improved yield of plants. The instant invention is more specifically explained by above examples. However, it should be understood that the scope of the present invention is not limited by the examples in any manner. It will be appreciated by any person skilled in this art that the present invention includes aforesaid examples and further can be modified and altered within the technical scope of the present invention.

While the foregoing written description of the invention enables one of ordinary skill to make and use what is considered presently to be the best mode thereof, those of ordinary skill will understand and appreciate the existence of variations, combinations, and equivalents of the specific embodiment, method, and examples herein. The invention should therefore not be limited by the above described embodiment, method, and examples, but by all embodiments and methods within the scope and spirit of the invention.

The invention claimed is:

1. A fungicidal combination consisting of:

mancozeb,
prothioconazole, and
benzovindiflupyr.

2. The combination as claimed in claim 1, wherein the combination is formulated into wettable powders, granules, dusts, soluble concentrates, suspension concentrates, oil in water emulsion, water in oil emulsion, emulsifiable concentrates, capsule suspensions, suspension concentrates combined with capsule suspensions, or oil dispersions.

3. A method of foliar application comprising applying the combination of claim 1 to a plant propagation material.

4. A method of controlling fungal diseases in plants comprising applying to a locus of the plant the combination of claim 1.

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